MACHINE LEARNING

Clear revision notes | Easy definitions | Readable formulas | Libraries | Viva questions

Reorganized topic-wise from the ML portions of your uploaded assignment write-up (Assignments 8-12). Formulas are written in plain text to avoid missing-symbol boxes.

How to read formulas: y_hat means predicted y; sum means add all values; sqrt means square root; x_bar means average of x.

1. Machine Learning basics

What is Machine Learning?

Machine Learning (ML) is a part of Artificial Intelligence in which a computer learns patterns from data and uses them to make predictions or decisions.

Type	Easy meaning	Examples in your practicals
Supervised learning	Learns from examples that already have correct answers (labels).	Linear Regression, Random Forest, KNN, SVM, Neural Network classifier.
Unsupervised learning	Finds patterns or groups in data without answer labels.	K-Means and Hierarchical Clustering.

Term	Meaning
Dataset	Complete collection of data used in a project.
Row / sample	One record, such as one customer or one email.
Feature (X)	Input column used to make a prediction.
Target (y)	Output column the model must predict.
Training data	Data used to teach the model.
Testing data	Separate data used to check the model on unseen examples.
Model	The learned pattern or algorithm.
Overfitting	Model learns training data too closely, including noise, and performs poorly on new data.
Underfitting	Model is too simple to learn the important pattern.

2. Data preprocessing

Data preprocessing means cleaning and converting raw data into a suitable format before using it in a machine learning model. It helps handle missing values, noise, inconsistent formats and categorical values.

Basic workflow

• Load and inspect the dataset (head(), shape(), info()).

• Find missing values and fill them or remove affected records when appropriate.

• Check unusual values and outliers before deciding how to handle them.

• Convert categorical/text values into numbers using encoding.

• Separate input features X from target y.

• Split data into training and testing sets.

• Scale numerical features when needed, then train and evaluate the model.

Train-test split

A common split is 80% training and 20% testing. The test data should be kept separate until evaluation.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)
```

• test_size=0.2 means 20% of the data is used for testing.

• random_state=1 makes the random split reproducible.

• stratify=y keeps class proportions similar in train and test sets (useful for classification).

Normalization and standardization

Method	Meaning	Formula
Normalization (Min-Max)	Usually scales values between 0 and 1.	x_scaled = (x - x_min) / (x_max - x_min)
Standardization (Z-score)	Makes mean 0 and standard deviation 1.	z = (x - mean) / standard_deviation

Important: fit the scaler using training data only, then use that same fitted scaler to transform test data. This prevents test-data leakage.

Feature scaling is especially important for distance-based algorithms such as KNN and for SVM. Random Forest generally does not require scaling.

3. Regression

Regression predicts a continuous numerical value, such as fare, price, salary or distance. Assignment 8 predicts Uber ride fare.

Linear Regression

A supervised algorithm that models a straight-line relationship between input feature(s) and a numerical target.

```
Simple: y_hat = b0 + b1*x
Multiple: y_hat = b0 + b1*x1 + b2*x2 + ... + bn*xn
```

y_hat = predicted value; b0 = intercept; b1...bn = learned coefficients; x1...xn = input features.

Random Forest Regression

A supervised ensemble algorithm. It builds many decision trees using different samples/features and combines their predictions. For regression, it usually averages the predictions from all trees.

```
Final prediction = (Tree 1 prediction + ... + Tree T prediction) / T
```

It can learn complex, non-linear patterns. More trees can make predictions more stable, but also increase computation.

Regression evaluation metrics

Metric	Meaning	Formula / interpretation
MSE	Mean Squared Error: average squared difference between actual and predicted values.	MSE = sum((y - y_hat)^2) / n
RMSE	Root Mean Squared Error; error is in the same unit as the target.	RMSE = sqrt(MSE)
R-squared (R2)	How much variation in the target is explained by the model.	R2 = 1 - [sum((y-y_hat)^2) / sum((y-y_mean)^2)]

For MSE and RMSE, lower is better. R2 closer to 1 generally indicates a better fit. R2 can be negative if the model performs worse than simply predicting the target average.

4. Classification

Classification predicts a category/class rather than a continuous number. Examples: spam/not spam, diabetes/no diabetes, customer leaves/stays.

Binary classification

A classification problem with exactly two possible classes. In the email practical: 1 = spam and 0 = not spam.

K-Nearest Neighbors (KNN)

A supervised, distance-based algorithm. It finds the K closest training examples to a new sample and assigns the class that appears most often among those neighbors.

```
Euclidean distance = sqrt((p1-q1)^2 + (p2-q2)^2 + ... + (pn-qn)^2)
```

• K is the number of neighbors considered.

• If K=5 and 3 neighbors are class A while 2 are class B, predict class A.

• Small K can be affected by noise; very large K can blur class boundaries.

• Feature scaling is important because KNN uses distances.

Support Vector Machine (SVM)

A supervised algorithm that finds a decision boundary (hyperplane) that separates classes with the largest possible margin. The nearest training points are called support vectors.

```
Decision boundary: w1*x1 + w2*x2 + ... + b = 0
```

SVM can use kernels to create non-linear boundaries. Feature scaling is commonly important.

Random Forest Classifier

Uses many decision trees and combines their class predictions, commonly by majority vote. It can be used for classification as well as regression.

Artificial Neural Network (ANN)

An ANN is a group of connected artificial neurons (nodes) arranged in layers. It learns by adjusting connection weights.

Part	Meaning
Input layer	Receives feature values.
Hidden layer	Processes inputs and learns patterns.
Output layer	Produces the final prediction.
Weights	Numbers controlling the strength of connections.
Bias	An adjustable value added to a neuron's calculation.
Epoch	One complete pass through the training dataset.
Activation function	Adds non-linearity so the network can learn complex patterns.

```
Neuron: z = (w1*x1 + w2*x2 + ... + wn*xn) + bias
Neuron output = activation(z)
```

Activation functions

Function	Formula	Meaning
Sigmoid	1 / (1 + e^(-z))	Output is between 0 and 1; often used for binary-classification output.
ReLU	max(0, z)	Negative values become 0; positive values stay unchanged.

Keras and TensorFlow

TensorFlow is a machine-learning/deep-learning framework. Keras is a high-level interface that makes it easier to create and train neural networks. The bank churn practical uses a neural network to predict whether a customer will leave.

5. Confusion matrix and classification metrics

A confusion matrix compares actual classes with predicted classes.

	Actual positive	Actual negative
Predicted positive	TP = True Positive	FP = False Positive
Predicted negative	FN = False Negative	TN = True Negative

- TP: predicted positive and actually positive.
- TN: predicted negative and actually negative.
- FP: predicted positive but actually negative.
- FN: predicted negative but actually positive.

Metric	Formula	Easy meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN)	How many total predictions are correct.
Error rate	(FP + FN) / Total	How many total predictions are wrong; equals 1 - accuracy.
Precision	TP / (TP + FP)	Of predicted positives, how many were truly positive?
Recall	TP / (TP + FN)	Of actual positives, how many did the model find?
F1-score	2 * (Precision * Recall) / (Precision + Recall)	Balances precision and recall.
Specificity	TN / (TN + FP)	Of actual negatives, how many were correctly identified?

ROC curve and AUC

The ROC curve plots True Positive Rate (recall) on the Y-axis against False Positive Rate on the X-axis at different thresholds.

```
TPR = TP / (TP + FN)
FPR = FP / (FP + TN)
```

AUC is the area under the ROC curve. A larger AUC generally means better separation between classes.

Null error rate

The error rate obtained by always predicting the majority class. It is a simple baseline for comparison.

For imbalanced datasets, accuracy alone can be misleading. Also check precision, recall, F1-score and the confusion matrix.

6. Clustering

Clustering is an unsupervised learning task that groups similar data points together. It can be used for customer segmentation, grouping documents and finding patterns when labels are unavailable.

K-Means clustering

K-Means divides unlabelled data into K clusters. Each cluster has a centroid (the mean position of its points). The algorithm assigns points to the nearest centroid and updates centroids repeatedly.

• Choose the number of clusters K.

• Initialize K centroids (the practical uses k-means++).

• Assign each point to its nearest centroid.

• Recalculate each centroid as the mean of its assigned points.

• Repeat assignment and update until the clusters stop changing significantly.

```
Centroid = (sum of all points in a cluster) / (number of points in that cluster)
```

WCSS / Inertia

Within-Cluster Sum of Squares measures how close points are to their own cluster centroid. K-Means tries to minimize this value.

```
WCSS = sum over clusters [sum of squared distances from each point to its centroid]
```

Elbow method

Run K-Means for several K values and plot K against WCSS. WCSS decreases as K increases. Choose the elbow point where the decrease starts becoming much smaller.

Hierarchical clustering

A method that builds a hierarchy of nested clusters. In agglomerative clustering, each point starts as its own cluster and the closest clusters are repeatedly merged. A dendrogram shows the merging process.

The uploaded practical title mentions K-Means and Hierarchical Clustering, but the write-up explains K-Means in more detail than hierarchical clustering.

7. Outliers, box plot, correlation and Haversine

Outlier

An outlier is a data point that is unusually different from most other observations. It may be an error or a genuine rare event, so investigate it before removing it.

Box plot and IQR

A box plot summarizes data using Q1 (first quartile), median (Q2), Q3 (third quartile) and whiskers. It helps identify possible outliers.

```
IQR = Q3 - Q1
Lower fence = Q1 - 1.5 * IQR
Upper fence = Q3 + 1.5 * IQR
```

Values outside these fences are commonly flagged as possible outliers, not automatically errors.

Correlation

Correlation measures the strength and direction of a relationship between two numerical variables. Pearson correlation ranges from -1 to +1.

• +1 = perfect positive linear relationship.

• -1 = perfect negative linear relationship.

• 0 = no linear relationship.

```
r = sum((x - x_mean)*(y - y_mean)) / sqrt(sum((x-x_mean)^2)*sum((y-y_mean)^2))
```

Correlation does not prove causation.

Haversine formula

The Haversine formula calculates the great-circle distance between two points on Earth using latitude and longitude. It can estimate pickup-to-drop-off distance in the Uber fare practical.

```
a = sin^2(Delta_lat/2) + cos(lat1)*cos(lat2)*sin^2(Delta_lon/2)
c = 2 * atan2(sqrt(a), sqrt(1-a))
Distance = Earth_radius * c (Earth radius is about 6371 km)
```

Use latitude and longitude in radians, not degrees.

8. Python libraries used

Library	What it is used for	Common examples
Pandas	Read, clean, select and manipulate table data.	import pandas as pd; read_csv(), read_excel(), head(), shape, drop()
NumPy	Numerical arrays and maths.	import numpy as np; array(), mean(), arange()
Matplotlib	Create charts and plots.	import matplotlib.pyplot as plt; plot(), scatter(), hist(), show()
Seaborn	Statistical plots; built on Matplotlib.	import seaborn as sns; heatmap(), boxplot(), scatterplot()
Scikit-learn	Preprocessing, ML algorithms, splitting and evaluation.	train_test_split, KNeighborsClassifier, SVC, LinearRegression, KMeans, metrics
TensorFlow	Framework for machine learning and neural networks.	import tensorflow as tf
Keras	High-level API for building neural networks.	from tensorflow import keras
SciPy	Scientific computing; hierarchical clustering tools.	scipy.cluster.hierarchy: linkage(), dendrogram()

Useful Scikit-learn classes and functions

Name	Purpose
train_test_split()	Split features and target into training and testing sets.
StandardScaler() / MinMaxScaler()	Standardize or normalize features.
LinearRegression()	Create a linear regression model.
RandomForestRegressor()	Create a random forest regression model.
RandomForestClassifier()	Create a random forest classification model.
KNeighborsClassifier()	Create a KNN classifier.
SVC()	Create a Support Vector Classifier.
KMeans()	Create a K-Means clustering model.
confusion_matrix()	Create the confusion matrix.
accuracy_score(), precision_score(), recall_score(), f1_score()	Calculate classification metrics.
mean_squared_error(), r2_score()	Evaluate regression predictions.
roc_curve(), roc_auc_score()	Calculate ROC curve points and ROC-AUC.
cross_val_score()	Evaluate a model over multiple folds.
GridSearchCV()	Search for good hyperparameter values using cross-validation.

9. Model improvement

Cross-validation

In K-fold cross-validation, data is divided into K parts (folds). The model is trained K times. Each time, one fold is used for validation and the other K-1 folds for training. The scores are averaged. Your practical refers to 5-fold cross-validation.

Hyperparameter tuning and GridSearchCV

A hyperparameter is a setting selected before/during training, such as K in KNN or the number of trees in Random Forest. GridSearchCV tests specified settings using cross-validation and reports the best settings and score.

fit(), predict() and score()

Code	Meaning
model.fit(X_train, y_train)	Train the model using training data.
model.predict(X_test)	Generate predictions for test inputs.
model.score(X_test, y_test)	Return the model's default score (often accuracy for classifiers and R-squared for regressors).

10. Practical-wise quick map

Assignment	Problem	Main ML topics
8	Predict Uber ride fare	Preprocessing, outliers, correlation, Haversine, Linear Regression, Random Forest Regression, MSE/RMSE/R2.
9	Spam vs not-spam email	Binary classification, preprocessing, KNN, SVM, train-test split.
10	Predict bank customer churn	ANN, Keras, TensorFlow, normalization, confusion matrix, accuracy.
11	Predict diabetes / no diabetes	KNN, train-test split, cross-validation, GridSearchCV, confusion matrix, precision, recall, ROC-AUC; objective also mentions Random Forest classification.
12	Customer segmentation	Unsupervised learning, K-Means, WCSS, elbow method; title also mentions hierarchical clustering.

11. One-page formula revision

Topic	Readable formula
Linear regression	y_hat = b0 + b1*x (simple); y_hat = b0 + b1*x1 + ... + bn*xn (multiple)
Min-Max normalization	x_scaled = (x - min) / (max - min)
Standardization	z = (x - mean) / standard_deviation
Euclidean distance	sqrt((p1-q1)^2 + ... + (pn-qn)^2)
MSE / RMSE	MSE = sum((y-y_hat)^2)/n ; RMSE = sqrt(MSE)
R-squared	1 - [sum((y-y_hat)^2) / sum((y-y_mean)^2)]
Accuracy	(TP+TN) / (TP+TN+FP+FN)
Error rate	(FP+FN) / Total = 1 - Accuracy
Precision / Recall	TP/(TP+FP) ; TP/(TP+FN)
F1-score	2 * Precision * Recall / (Precision + Recall)
IQR fences	IQR=Q3-Q1; lower=Q1-1.5*IQR; upper=Q3+1.5*IQR
K-Means WCSS	Sum of squared distances from points to their cluster centroids
Sigmoid / ReLU	Sigmoid=1/(1+e^(-z)); ReLU=max(0,z)

12. Short viva questions and answers

What is data preprocessing? Cleaning and converting raw data into a suitable form for machine learning.

What is a feature? An input variable used by a model to make a prediction.

What is a target? The output value or class the model must predict.

What is regression? Predicting a continuous numerical value.

What is classification? Predicting a category or class label.

What is binary classification? Classification with two possible classes.

What is an outlier? A value unusually different from most other observations.

What is an ANN? A network of connected artificial neurons arranged in layers.

What is a confusion matrix? A table comparing actual and predicted classes.

What is KNN? An algorithm that predicts a class using the majority class among K nearest samples.

What is SVM? A classifier that finds a separating boundary with a large margin.

What is Random Forest? An ensemble of decision trees whose predictions are combined.

What is clustering? Grouping similar data points without predefined labels.

What is K-Means? An algorithm that partitions data into K groups around centroids.

What is the elbow method? A method to choose K by finding where the WCSS curve starts flattening.

What is normalization? Rescaling values to a common range, often 0 to 1.

What is standardization? Rescaling values to mean 0 and standard deviation 1.

What is overfitting? When a model learns training details/noise and performs poorly on new data.

What is cross-validation? Training and validating across different data folds.

What is GridSearchCV? A tool that tests hyperparameter combinations using cross-validation.

What is precision? Among predicted positives, the proportion that are truly positive.

What is recall? Among actual positives, the proportion correctly found.

What is RMSE? Square root of mean squared prediction error, in the target's unit.

What is R-squared? A regression score describing the variation in the target explained by the model.

What is Haversine? A formula for distance between two Earth locations using latitude and longitude.

13. Exam reminders

• Regression = numerical output; classification = category output; clustering = discover groups without labels.

• Separate X (features) and y (target) before training.

• Evaluate using unseen test data, not only training data.

• Scaling matters for KNN and SVM; it is usually not needed for Random Forest.

• Lower MSE/RMSE is better; higher R-squared is generally better.

• Know TP, TN, FP and FN before learning classification metrics.

• For K-Means, remember K, centroid, WCSS/inertia and elbow method.

• An outlier is not automatically a mistake; check its context.

Source note: Core practical topics are drawn from Assignments 8-12 in your uploaded ML assignment write-up. The formulas are presented in plain ASCII notation for readability; brief explanations are included to make the notes easier to study.